

Jugal Gajjar

812jugalgajjar@gmail.com | github.com/JugalGajjar | linkedin.com/in/jugal-gajjar | jugalgajjar.github.io | Google Scholar

RESEARCH SUMMARY

My research focuses on trustworthy, neurosymbolic and reasoning-centric systems, particularly at the intersection of large language models (LLMs), agentic workflows, and explainable AI (XAI). I am interested in developing structured and verifiable AI systems that combine neural representations with symbolic reasoning, graph-based knowledge, and execution-grounded validation for reliable decision-making, business and scientific automation, and code intelligence.

EDUCATION

M.S. in Computer Science (Machine Intelligence & Cognition)

The George Washington University, *Washington, DC*

GPA: 3.88/4.0

May 2026

B.Tech. in Computer Science and Engineering

Navrachana University, *Vadodara, India*

CGPA: 9.39/10

May 2024

RESEARCH PUBLICATIONS

RSAT: Structured Attribution Makes Small Language Models Faithful Table Reasoners

First Author; SURGeLLM Workshop (Proceedings Track) at ACL 2026

Jul. 2026

Feature Attribution Stability Suite: How Stable Are Post-Hoc Attributions?

Co-Author; XAI4CV Workshop (Proceedings Track) at CVPR 2026

Jun. 2026

Bridging Semantics & Structure for Software Vulnerability Detection using Hybrid Network Models

First Author; 14th International Conference on Complex Networks and their Applications

Dec. 2025

HyperComplEx: Adaptive Multi-Space Knowledge Graph Embeddings

First Author; 2025 IEEE International Conference on Big Data (IEEE BigData 2025)

Dec. 2025

SecureFixAgent: A Hybrid LLM Agent for Automated Python Static Vulnerability Repair

First Author; 2025 International Conference on Machine Learning and Applications (ICMLA)

Dec. 2025

MLCPD: A Multi-Language Code Parser Dataset with Universal AST Representation

First Author; arXiv preprint (cs.SE)

Oct. 2025

MalCodeAI: Autonomous Vulnerability Detection and Remediation via Language Agnostic Code Reasoning

First Author; IEEE 26th International Conference on Information Reuse and Integration (IRI 2025)

Aug. 2025

PATENTS

A Portable Scanning and Computing Device

Co-Inventor; Indian Patent Application No. 202321083926

Jan. 2025

PROFESSIONAL SERVICE

Program Committee Member:

- 38th IEEE International Conference on Tools with Artificial Intelligence (ICTAI)

Nov. 2026

Reviewer:

- ICMLA 2026

- ACL 2026 (GEM Workshop, SURGeLLM Workshop, CDL Workshop) Jul. 2026
- IEEE USBEREIT 2026 May. 2026
- ICMLA 2025 (ROSE-LLM Special Session) Dec. 2025

SELECTED RESEARCH PROJECTS

Cross-Language Vulnerability Detection, Validation, and Repair (M.S. Thesis) Jun. 2025 – Mar. 2026

- Built a uAST-based agentic framework for vulnerability detection and repair, achieving a 69.74% end-to-end resolution rate on a 120K-sample dataset.
- Designed a GNN + transformer fusion model, reaching 74.4–78.2% zero-shot cross-language F1.
- Developed an end-to-end pipeline with execution-based validation, reducing false positives by up to 62% and successfully repairing 81–87% of cases.

RSAT: Structured Attribution for Faithful Table Reasoning in Small LLMs Dec. 2025 – Mar. 2026

- Designed a two-phase SFT + GRPO training pipeline enabling small language models (1–8B) to generate step-by-step reasoning with explicit cell-level citations.
- Developed a composite reward integrating NLI-based faithfulness, citation validity, and parsimony for structured reasoning.
- Achieved $3.7\times$ improvement in attribution faithfulness over supervised fine-tuning baselines.

VulnGraph: Graph + LLM Embeddings for Vulnerability Detection Jul. 2025 – Sep. 2025

- Engineered a multimodal fusion model integrating AST/CFG graph embeddings with LLM-based semantic representations.
- Integrated structural program representations (AST/CFG) as symbolic constraints to guide neural-LLM feature fusion via a two-way gating mechanism for effective alignment.
- Achieved 93.57% accuracy, outperforming GNN-only (+8.36%) and LLM-only (+17.81%) baselines with interpretable outputs.

HyperComplex: Adaptive Multi-Space Knowledge Graph Embeddings Apr. 2025 – Oct. 2025

- Proposed a hybrid knowledge graph embedding framework combining hyperbolic, complex, and Euclidean spaces through relation-adaptive attention mechanisms.
- Developed a scalable multi-space model with consistency regularization for heterogeneous relational reasoning.
- Achieved 0.612 MRR on a 45M-triple knowledge graph with near-linear scalability across heterogeneous relations.

FASS: Feature Attribution Stability Suite for Robust XAI Evaluation Nov. 2025 – Feb. 2026

- Developed a benchmark for evaluating attribution stability under prediction-invariant perturbations (geometric, photometric, compression).
- Introduced a three-axis stability decomposition: structural similarity, rank correlation, and top- k overlap.
- Enabled systematic comparison of post-hoc explanation methods for robustness and reliability analysis.

TECHNICAL SKILLS

Programming: Python, C++, Java, SQL, Bash

Machine Learning & AI: PyTorch, TensorFlow, Hugging Face Transformers, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Reinforcement Learning from Human Feedback (RLHF), Graph Neural Networks (GNNs), AI Agents

Systems & Infrastructure: Linux, Git, Docker, Apache Spark, Hadoop, AWS, Google Cloud

Research Interests: Neurosymbolic AI, LLM Reasoning & Planning, Explainable AI (XAI), Knowledge Graph Reasoning, Agentic AI Systems, AI for Scientific Workflows, Software Security & Program Analysis

RELEVANT EXPERIENCE

Graduate Assistant

The George Washington University, *Washington, DC*

Sep. 2025 – Dec. 2025

- Supported an upper undergrad- & graduate-level Big Data and Analytics course (CSCI 4907/6444).
- Guided and graded students on assignments and projects, focusing on tools like Spark, Hadoop, and S3 Bucket, and programming in languages like Python, Java, and Linux-Bash.

- Assisted with analytical methods, including machine learning, and provided support for cloud-based and distributed systems projects.

Teaching Assistant

Navrachana University, *Vadodara, India*

Jan. 2024 – May 2024

- Mentored undergraduate students through code review, grading, and structured feedback on 750+ lab reports.
- Assisted in designing rubrics, evaluating technical writing, and improving student understanding of test-driven development.
- Supported curriculum delivery along with faculty, focusing on code quality, documentation, and debugging.

NVIDIA Jetson AI Project Coordinator

Navrachana University, *Vadodara, India*

Mar. 2022 – Jun. 2022

- Conducted workshops on CV and NLP using the Jetson Nano platform, introducing students to embedded AI and real-time inference.
- Mentored 5+ student research projects selected by the NVIDIA Deep Learning Institute.
- Promoted student research initiatives and hands-on learning in edge AI systems.

CERTIFICATIONS & HONORS

Sustainability Award (Complex Networks 2025 Conference)	2025
NVIDIA-Certified Associate: Generative AI LLMs	2025
GW SEAS Dean’s Award Scholarship	2024
Runner-Up, Tinkerthon 2.0 Hackathon	2024
SSIP Innovation Grant (Project EzyCart)	2023
NVIDIA Jetson AI Specialist	2022